الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

UGV-Based Multi-Spectral Detection of Surface & Shallow-Buried Landmines

AI-Powered Multi-Spectral Rovers for Humanitarian Demining



SUPERVISORS:
Dr. Riad Sonbol | Eng. Marwa Alsafadi | Eng. Alhussain Tofailea

STUDENTS:
Leen Alhomsi | Omar Alkabbani | Firas AlSowis

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The Project Concept & Background Statement

PROJECT CONCEPT

- Developing autonomous Unmanned Ground Vehicles (UGVs) specialized in surface and shallow landmine identification.
- Employing multi-spectral optical and sensor fusion
- Providing rapid, high-accuracy spatial mapping



Hardware & AI Baseline Benchmark Analysis

Feature Category	MineInsight Benchmark (2025)	SULAND Framework (2024)	UAV-Based Detection (2020)
Robotic Platform	Clearpath Husky A200 UGV + Universal Robots 6-DOF Manipulator Arm	UGO-first / Clearpath Jackal UGV (and compatible UAV survey configs)	DJI Matrice 600 Pro & Phantom 4 Pro UAVs
Hardware & Sensors	Dual LiDARs (Livox Mid-360 & AVIA), RGB, VIS-SWIR, LWIR Thermal, IMU, RTK-GPS	iPhone 13 Pro (RGB + LiDAR suite), RTK-GPS module, holographic GPR radar	Parrot Sequoia Multispectral Camera (RGB, Red Edge, NIR) & Thermal IR
AI Models & Software	3D LiDAR SLAM (SLICT), camera-LiDAR targetless calibration, AprilTag auto-labeling	YOLOv8 (ONNX nano & small models) real-time browser inference	Faster Regional-CNN (Faster R-CNN) & Pix4DMapper orthophotos
Datasets & Targets	15 AP/AT landmine types + 20 clutter objects across off-road dirt tracks	PFM-1 'Butterfly' and PMA-2 'Starfish' mines across grass/gravel	PFM-1 scatterable antipersonnel mines across grass & rubble

Project Vision & Proposed Solution

- ✦ Autonomous Robotic Deployment
- ✦ Real-Time Deep Detection Model
- ✦ Multi-Modal Sensor Integration
- ✦ Target Mine Focus (PFM-1 Butterfly)



System Workflow & Methodology

AI & COMPUTER VISION WORKFLOW

1. Paper Analysis
MineInsight & PFM-1 research review

2. Dataset Analysis
MineInsight dataset deep dive

STEP 3 (ACTIVE)
Model Replication
Fine-tuned YOLOv8 & CV

4. Benchmarking
Results comparison with original

5. Fusion Pipeline
YOLOv8 + CV feature fusion

6. Deployment
Simulated environment testing

ROBOTICS & CONTROL SYSTEM WORKFLOW

1. Literature Review
Robot & sensor selection papers

2. Platform Choice
Robot design & environment selection

STEP 3 (ACTIVE)
ROS 2 & Gazebo
Simulation & SLAM

4. SolidWorks CAD
Chassis design & URDF export

5. AI Integration
Deploy CV model in simulation

6. Evaluation
Test performance & motion control

Progress Dashboard

AI & Computer Vision Stream

Step 3 (Model Replication & Frame Inspection)

30% Phase Completion

Baseline Replication	Data Inspection	Label Transition
YOLOv8 + Classical CV baseline active	Deep track frame and terrain analysis	Moving to time-based target labels

Robotics & Control System Stream

Step 3 (ROS 2 Jazzy & Gazebo Integration)

30% Phase Completion

Simulation Stack	SLAM & Mapping	Terrain Testing
ROS 2 Jazzy workspace & Gazebo nodes	Point-cloud & 3D spatial mapping active	Physics simulation on rough surfaces

Technical Specification

Selected Item / Subsystem	System Role
Selected Environment	Off-Road Terrain - Forest
Chassis & Mechanical Frame	Mechanical Structure
Tracks & Motors	Propulsion & Drive
Power Supply	Power Management
Robotic Arm	Manipulation Subsystem
Raspberry Pi	Edge Processing
Microcontrollers	Embedded Control
Lidar	Spatial Perception
Camera	Computer Vision

